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Project Title: **Portable Cooling Pad: A refrigeration Cycle for Laptop**

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Abstract

With advancement of Technology laptops are frequently used for intensive tasks such as gaming, video editing, machine learning, simulation and CAD design. As a result overheating in laptops becomes an inseparable part of the users. This project introduces an innovative solution: a portable cooling pad utilizing the refrigeration cycle to effectively regulate and reduce laptop temperatures. The traditional cooling pads rely solely on passive airflow. But this device incorporates a compact heat exchanger to achieve efficient and sustained cooling using the refrigeration cycle. This cooling pad, comprising a compressor, condenser, expansion valve, and evaporator, actively dissipates heat generated by the laptop. Thus it maintains an optimal operating temperature. While designing this project portability, energy efficiency, enhancing the longevity and performance of laptops were the key concerns. Prototyping and testing demonstrated a significant reduction in surface temperature compared to conventional cooling methods. Thus it became a viable solution to ensure thermal management in portable computing devices.

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Chapter 1: Introduction

1.1 Introduction to Refrigeration Cycle for Laptop Cooling

Overheating has recently become one of the major concerns in laptop design mainly because of higher technological development and the increased need for more powerful laptops. Although laptops contain powerful processors they are designed to perform complex computations and due to their compact size, interior ventilation is severely restricted. Conventional approaches for the item include fan-based air cooling, or passive cooling pads, which can at best only restrain the issue, by improving the rate of flow of air, or offering some form of cooling. Unfortunately, these are not enough for overcoming the heat that is produced during intensive and protracted use with little opportunity for external cooling.

The refrigeration cycle, which has become one of the many thermal management models in the field of thermal engineering, is becoming an ideal solution for efficient thermal management as well. Common to air conditioners and refrigerators the refrigeration cycle is a heat exchange process in which heat is transferred from a low temperature zone to a high temperature zone in a closed circuit. Some of the major parts of this cycle consist of a compressor for increasing pressure of the refrigerant, a condenser for draining the heat, an expansion valve to decrease pressure and an evaporator to absorb heat. By incorporating this cycle into a cooling pad which can be transported from place to another, laptops can obtain active heat elimination and therefore improve their thermal efficiency. The purpose of this project is to use information from previous related projects and make improvements to the refrigeration cycle to incorporate in a small scale portable unit for a laptop computer.

1.2 Problem Statement

In the engineer's world especially the Mechanical engineer, the laptops are more than just personal gadgets; they are work instruments and innovation tools. Laptops are often used by engineers to perform simulations, and 3D modeling, design optimization involving software such as ANSYS , SolidWorks, AutoDesk Inventor, and MATLAB. These programs require outsized amounts of computational power and, consequently, cause processors and GPUs to run flat out for longer than ideally desired. Consequently, the parts and elements of the system produce a lot of heat and, therefore, can overheat.

When laptops overheat, mechanical engineers face a number of issues. First, it hinders performance where because of heating, the system will slow down to avoid such damage. Such can lead to slow simulation runs, wrong computation, and a decrease in efficiency of processes. Second, high temperatures impact internal hardware, which decreases a laptop's durability and consistently requires repairs or replacement, which is unsuitable since laptops are a professional's own money. Last but not the least, since Laptops also contain processors, they get hot degrees, especially in areas with no air conditioning, making it Hard to concentrate.

Previous cooling methods like fans and air cooling mats cannot take the heat of the thermal load that arises during complex engineering processes. These solutions may bring temporary relief but do not solve the problem of overheating. This leaves the need for a more enhanced, effective, and transportable cooling style for particular needs in mechanical engineers or any avid laptop user.

1.3. Objectives of the Project

This project will tackle the issue in utilizing an improved cooling pad on the laptop through the use of refrigeration cycle. The objectives are as follows:

Design and Development: In order to achieve an active cooling system for laptops through the fabrication of a portable cooling pad inspired by the principles of the refrigeration cycle. It is desired that the components of the system which include the compressor, the condenser, the expansion valve and the evaporator should be compactly designed and should be portable.

Performance Enhancement: For the better solution in thermal issues of laptop and to maintain a steady performance during the usage of demanding application software for a longer period of time. The cooling pad is intended to keep the surface of laptops at safe working temperature regardless of the amount of work it is involved in.

Energy Efficiency: In order to make the cooling pad as energy efficient as possible as to make it more practical to use in everyday life.

Portability and Usability: In this regard the goal was to design and develop a portable device that mech. Engineers or any other professional can easily transport and utilize without much hassle in terms of setting up the environment.

Testing and Validation: In the effectiveness of the cooling pad experiment as a way of determining efficacy, compared to conventional cooling techniques. The control factors are the amount of heat, overall energy usage, noise, and the level of satisfaction that users have on the building.

Longevity and Reliability: To increase the lifespan of laptops by delivering an efficient cooling system that can negate the straining effect of heat on laptop hardware thus reducing costs for users.

By doing this project, the refrigeration cycle is going to be modified and improved for entirely new use, which is going to offer a fresh concept to one of the most powerful and yet enduring problems in portable computers. The ultimate goal is to provide the mechanical engineers and other professionals with the tool that will increase their effectiveness in their job, but at the same time safeguard their significant resources and their devices.

Chapter 2: Literature Review

2.1 Literature Review

Modern laptops generate significant heat due to increasing computational power, necessitating advanced cooling mechanisms to maintain performance and device longevity. The exploration of refrigeration cycles offers a novel solution to address thermal management challenges in compact electronic systems.

Conventional Cooling Methods

Traditional laptop cooling systems include passive methods, such as aluminum heat sinks, and active methods, like fans and heat pipes. These approaches often struggle with efficiency under heavy workloads, failing to meet the demands of modern high-performance laptops [1].

Innovative Cooling Technologies

Advancements include thermoelectric cooling, phase change materials, and refrigeration cycles. Thermoelectric coolers (TECs) rely on the Peltier effect but are limited by high energy consumption. Phase change materials provide passive cooling but lack sustainability under extended use [1]. Refrigeration cycles, with their active heat removal capability and consistent temperature control, show the most promise [3].

Refrigeration Cycles in Laptop Cooling

Refrigeration cycles use compressors, condensers, evaporators, and expansion valves in a closed-loop system. Lee and Yoon's study demonstrated their feasibility, achieving substantial temperature reductions in notebook computers, outperforming traditional cooling methods [3].

R-134a Refrigerant Performance

R-134a's high latent heat and thermal conductivity make it ideal for small-scale refrigeration. Studies confirm its efficiency in heat transfer and environmental compliance, making it a preferred choice for laptop cooling applications [4][6][7].

Optimization of Condenser and Evaporator

Efficient condenser and evaporator designs are crucial for enhancing performance. Research highlights the role of copper tubing for its high thermal conductivity and capillary tube configurations for balanced heat exchange efficiency [8][9].

Challenges and Future Prospects

Key challenges include component miniaturization, energy efficiency, and integration into

compact chassis. Advances in microchannel heat exchangers and environmentally friendly refrigerants offer potential solutions [1][6][7]. Further research is needed to address these limitations and enhance sustainability.

In conclusion, integrating refrigeration cycles into laptop cooling represents a transformative advancement. Leveraging R-134a and optimized component designs offers a practical solution to heat dissipation challenges in high-performance laptops. Future efforts should address scalability and sustainability for broader adoption of this technology.

Chapter 3: Overview of the Project

3.1 Key Features

1. **Active Heat Removal**

This differs from passive cooling techniques under which the heat finds its way to the laptop surface and or other parts of the computer. This is accomplished by using the heat transfer through the means of refrigerant, preferably a closed circuit.

2. **High Cooling Efficiency**

The cycle basically comprises a compressor, a condenser, an expelling valve and an evaporator, all work in unison to ensure low temperature conditions are maintained. It can produce lower temperatures than the other conventional fans or non-electric cooling pads.

3. **Compact and Portable Design**

The system is designed in the form of a cooling pad which is, therefore, easily transportable and very light in weight suitable for users on the move. The blocks are shrunk down into ameleri reduced form factor without saving performance.

4. **Energy Efficiency**

The system is then designed to be energy efficient just to extends its capabilities without stressing power sources. This makes it useful for those that use laptops and are worried about power consumption.

5. **Compatibility**

Cooling pad ensures it can cool all types of laptops because the manufacturers create them to accommodate all kinds of laptops in the market.

6. **Enhanced Thermal Regulation**

Due to the stable working temperature, the cooling pad avoids the modulation of speed that occurs when a notebook is running high demanding programs such as gaming, video editing, or executing simulation applications.

7. **Prolonged Laptop Lifespan**

In essence, the heat dissipation brought by the system helps to reduce the strains put on more inside elements of the laptops; this in one way contributes to increasing the overall life expectancy of the laptops.

8. **Quiet Operation**

The refrigeration system works with less noise than conventional fan-cooling solutions and is less obtrusive for users in noise-sensitive environments.

9. **Minimal User Setup**

The cooling pad can actually be used immediately through a plug and play system. Customers can lay their laptops directly on the pad and switch it on and cooling begins right away.

10. **Flexibility to Work with Different Loads of Heat**

This allowed the system to be versatile enough in that it could take the amount of heat generated thereby optimizing its performance in different tasks or temperature settings.

11**. New Idea for Heat Exchanger Configuration**

It contains a rather compact heat exchanger, which is responsible for efficiently transferring heat to the refrigerant, thus increasing cooling capacity of the pad.

12. **Reduction in Noise Levels**

They are optimally designed to produce less noise than enclosures featuring fan blowers, more acceptable in the workplace or studying.

13. **Environmental Considerations**

The cooling pad may utilise eco-friendly refrigerants which are in accord with the GlobalReduce greenhouse gases emission standards = sustainability.

3.2 Operating Principles of a Refrigeration Cycle

Compression (Compressor)
The cycle commences with a low pressure and low temperature gaseous refrigerant. The compressor raises the pressure and the temperature of the refrigerant so that it becomes high pressure vapor. This process also needs energy input usually in the form of electricity input The process also involves the input of energy which is usually electricity. High pressure high temperature refrigerant gas is then passed into the condenser for the further stage of its cycle.

Heat Rejection (Condenser)
The hot high pressure refrigerant gas enters the condenser, usually a condenser coil or a series of fins exposed to the surrounding air. In this case, the refrigerant heat transfers to the surroundings through heat exchange with airflow around the condenser. Through cooling the refrigerant changes to a high pressure liquid but it carries most of the energy in the form of latent heat.

Expansion (Expansion Valve)
The high-pressure liquid refrigerant then falls through the expansion valve (or a throttling device). There is a notable decrease in temperature during this process due to a largely reduced pressure within the refrigerant, from the compressor. This leads to a situation where it is flooded with low pressure liquid as well as vapor concurrently. Since it is in the low pressure now the refrigerant is returned to the evaporator where it absorbs the heat from the object to be cooled.

Heat Absorption (Evaporator)
The low pressure refrigerant then moves into the evaporator; this is placed in thermal connection with heat load for example a laptop in your car. When it actually absorbs heat from the laptop for instance, the refrigerant vaporizes from a low pressure liquid to a low pressure gas. He explained that during this phase change the material is capable of absorbing heat and thus cooling the surface and interior of the laptop.

Cycle Repeats
The low pressure refrigerant gas cycles back to where it begins the process, the compressor. Both the constant circulating of the refrigerant and the repetitive absorption and rejection of heat keep up the cooling process.

Key Points of the Refrigeration Cycle:

Thermodynamics: The cycle works thermodynamically, especially heat exchange between a cooler body (laptop) to a hotter body (normal environment).

Refrigerant: A working fluid characterised by specific thermal parameters is employed for a heat transfer in the cycle.

Closed System: The process occurs in a cycle of its own where there is no leakage of the refrigerant and a maintained constant cooling temperatures.

Active Cooling: While passive cooling strategies work by halting the accumulation of heat around the laptop, the refrigeration cycle involves actively removing heat, hence providing more efficient cooling.

This process allows constant removal of heat making it useful in high end laptops that are bound to produce a lot of heat.

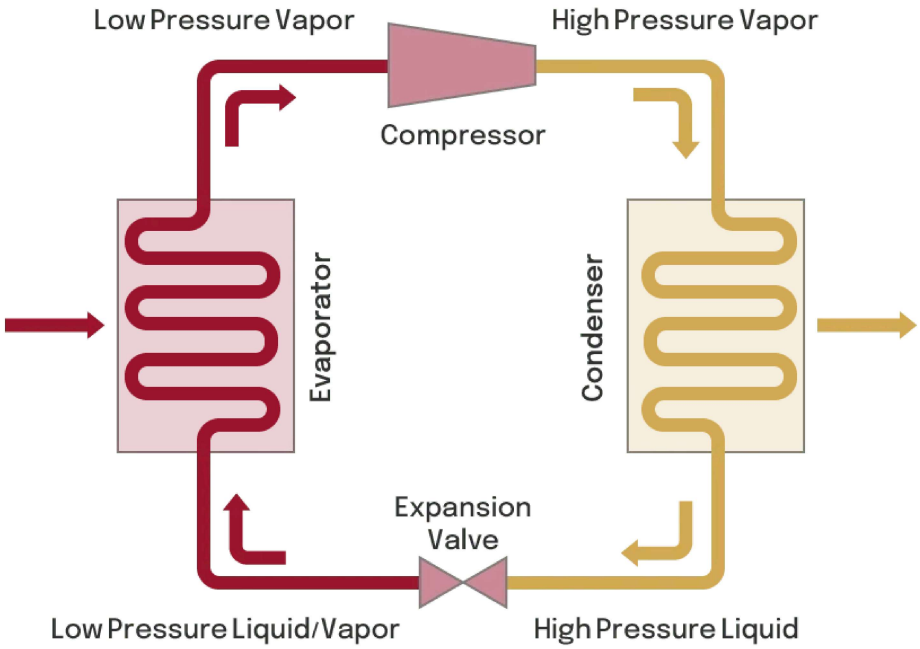


Fig: Refrigeration Cycle

3.3 Advantages

1. Superior Cooling Performance
The refrigeration cycle actively cools through absorption and elimination of heat and the internal temperatures of laptops connected with the cooling pads are relatively lower than using an ordinary cooling pad or fan. It also removes the issues with noise, which in turn helps to get the best out of the CPU regardless of the pressure it receives, be it gaming, video editing or conducting complex engineering simulations.

2. Use of R-134A Refrigerant
R-134A is one of the best known new generation non-toxic, non-ozone depleting refrigerants with excellent thermo physical properties. It has properties that do not set it alight and a relatively low level of emissions rating in comparison to the previous coolant compound, R-22, and is preferable for use in portable air conditioning systems. In small-scale usage, such as in your cooling pad, it effectively and safely conveys heat operations by means of its efficiency.

3. High enduring and stress holding capacity
Your system can handle hydraulic pressure and up to 200 psi which makes your system very reliable and strong even when confronted with such conditions. Such a level of durability defines your cooling pad appropriate whenever enhanced effectiveness, reliability, and durability are required in line with unrelenting operational needs.

4. Improved Thermal Stability
If you design your cooling pad to work based on the refrigeration cycle, then you can avoid problematic changes in temperature which may cause thermal throttling or even ruin your hardware. This is even more important when used in high I/O devices, such as high-performance laptops with complex software for extended usage.

5. Portability and User-Friendly Design
This type of cooling pad is portable and easy to use in spite of a highly developed refrigeration system that has been designed for it. It does make it a viable solution for anyone who needs performance cooling but doesn’t want to sacrifice portability for it; professionals, gamers, engineers etc.

6. Increased Laptop Longevity
Overheating is one of the significant issues that may harm the devices and leads to the destruction of a laptop’s overall parts such as the CPU, the GPU and motherboards . Thus, your cooling pad lowers stress levels on these components, hence an increased manufacturer’s life of the laptop.

7. Quiet Operation
The refrigeration cycle is also less noisy as compared to the system with high fans in cooling, which creates less disturbance to the user, particularly in professional or quiet scenes.

8. Energy Efficiency
Consequently, the current consumption is optimized according to the minimum value required to produce a maximum cooling rate. It is a low energy cooling pad because efficient components complement the R-134A refrigerant and the pads require a low power for operation, hence eco-friendly.

9. Adaptability of Systems for Extreme Conditions
Hydraulic pressure up to 200 psi shows us that this system can be used in varied conditions that are either high pressure or even industrial environments thus becoming highly functional and safe.

10. Improved Performance for Professions
The cooling pad is especially useful for individuals working in their careers such as mechanical engineers, gamers or those which involve the use of laptops in developing content by running frequent complex applications. Superior cooling capabilities keep everything running evenly and without hitch during workload activities such as simulation, rendering, and multithreading.

11. The fresh approach to using refrigerators in portable devices.
Your project involves using the refrigeration cycle, a system mostly implemented on massive systems like fridges, and air conditioners to create a laptop cooler; something that initially, cannot be changed but shows creativity as well as the capability to revolutionize laptop cooling solutions.

12. Eco-Friendly Solution
The requirement of R-134A refrigerant corresponds to global trends for minimizing the ecological risk of conditioning systems. It compels to meet contemporary environment standards while producing a high cooling effect.

3.4 Limitations and Possible Solutions

1. Complexity of Design

Limitation: Refrigeration cycle consists of four elements (compressor, condenser, expansion valve, evaporator), it is important to connect and install all of them properly. This makes the design more challenging than the conventional cooling pad kind of products.

Possible Solution: The system can be simplified by utilising compact pre-assembled refrigeration modules or miniature sub-assembly components. This is good because modular designs enable the assembly line and the process of maintenance to be less complex.

2. Higher Cost

Limitation: Refrigeration systems are costlier in production because of component costs as seen with compressors and refrigerants.

Possible Solution: This is done by using cheap and ubiquitous parts to create a system. Additional savings could be achieved by expanding cooperation with component providers. Furthermore, positioning the system as a product for the experts and gamers may help cover the costs.

3. Increased Power Consumption

Limitation: Compressor assembly along with other accessories this system requires much more power than a fan based cold air system.

Possible Solution: Optimize the usage of energy efficient compressor and novel refrigerant HFOs as well to minimize energy usage. By having smart sensors work for controlling the cooling depending on the thermal loads, constant energy consumption is reduced.

4. Size and Weight

Limitation: Decomposition of a cooling pad into its refrigeration components is conspicuous because it reduces portability due to increased mass.

Possible Solution: Downscale the system with small micro-compressors, and small heat exchangers. Engadget materials including alumina alloys or composites could mean lesser overall straight weight without compromising durability.

5. Noise Levels

Limitation: Compressor and refrigerant flow source creates noise and this may prove to be inconvenient to users in quiet conditions.

Possible Solution: The equipment should be mounted on low noise oil flood compressors, and advanced vibration damping products. Without considering the latest designs of airflow for cooling, identifying fans that produce minimal noise is also possible.

6. Refrigerant Leakage

Limitation: Refrigerants containing compounds such as R- 134A can escape from the system leading to reduced efficiency and the environment affected.

Possible Solution: Use a sealed system, which has complex leak-detection arrangements. The deleterious effects can however be contained by switching to more environmental friendly refrigerants with low global warming potential (GWP).

7. Reduced Operational Productivity in Special Circumstances

Limitation: The refrigeration cycle may be affected at very high temperatures of the surroundings in that the condenser may not perform heating rejection well.

Possible Solution: Optimise the flow of air across condensers or position additional thermoelectric coolers as supplementary systems for low temperatures.

8. Maintenance Requirements

Limitation: Interests may include checking on frosts, re-establishing refrigerant charges and as well checking on the performance of the compressor.

Possible Solution: Employ refrigeration units which do not need to be maintained and are covered. Provide self-diagnosing features with lights for refrigerant and the performance of individual components for the users.

9. Environmental Concerns

Limitation: Currently used refrigerants, such as R-134A, are more environmentally friendly than earlier refrigerants, but still contain a GWP.

Possible Solution: Go for the currently available fresh refrigerants like R-1234yf or hydrocarbons which comprises negligible GWP. They are environmentally friendly and they also meet modern day regulations.

10. Compatibility Issues

Limitation: The system of cooling can in some way be incompatible with all the designs of laptops especially if varying in size and in dissipation of heat.

Possible Solution: Make the cooling pad in such a way that it has mechanical flexibility, so one can adjust the angle and position, or use the pad in different configurations, the modular parts should fit all types of laptops and their heat production.

11. Heat Loss To the Environment

Limitation: The refrigeration cycle takes heat from the laptop, then expels this heat to the surrounding, which makes a difference in comfort within small enclosed areas.

Possible Solution: Modify the aspects of condensers in terms of the design of the fins that can be more efficient, and fans that make low noise while removing heat. On the other hand, one might want to introduce some enthusiast system to push the hot air away from the user.

12. Unavailability of 50W Compressor

Limitation: 50W compressor is unavailable in the market.

Possible Solution: The compressor can be custom made or DC compressor can be used instead of 50W compressor.

13. Startup Time for Cooling

Limitation: After some few moments a refrigerator system may take time to cool to an optimal level.

Possible Solution: Outline pre-charged refrigerants which are also referred to as systems with very short warm-up periods. On the other hand, add a partial cooling assist fan system prior to the full refrigeration cycle for condenser stabilization.

Summary of Solutions

By addressing these limitations through innovative design and technology, refrigeration-based laptop cooling pads can become a practical, efficient, and widely accepted solution. The focus should be on:

- Miniaturization and modular design.
- Energy efficiency and eco-friendly refrigerants.
- User convenience and compatibility.
- Robust, maintenance-free, and durable systems.

Chapter 4: Design

4.1 Design Factors Considered

Process Requirements: To cool laptop from 44°C to 30°C by using ambient air flow of 2.51 m/s at 20°C

Dimensions:

Evaporator:

Length: 5.56m
Diameter: 10 mm
Thickness: 1 mm
Material: Cu

Condenser:

Length: 2.78m
Diameter: 10 mm
Coolant: R-134A
Mass flow rate: 2.93 m/s

Expansion Valve:

Flow capacity: 10-12 kg/h
Orifice size: 0.5-1 mm
Refrigerant: R-134A

4.2 Design Drawings

Software used: SolidWorks



Fig 1: Solidworks Drawing of designed model

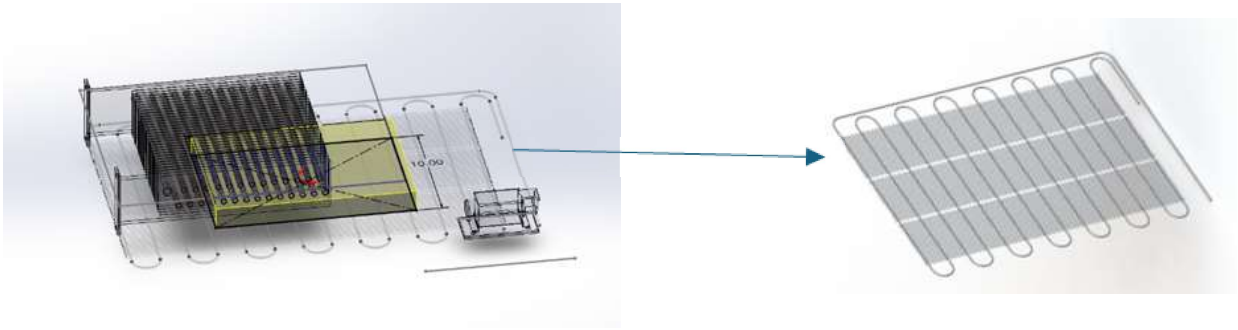


Fig 2: Condensar

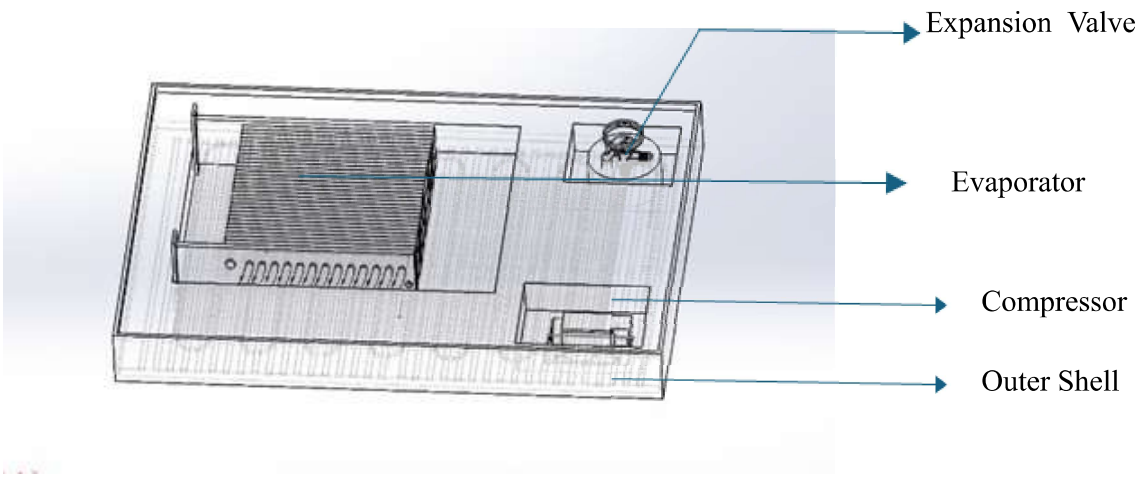


Fig 3: Position of differents parts

4.3 Fabricated Project



Fig: Implemented Laptop Cooler

Chapter 5: Mathematical Modelling and Calculations

5.1 Software Calculations

```
import math

# Constants
R134A_PROPERTIES = {
    "thermal_conductivity": 0.08,  # W/m-K
    "specific_heat": 1420,        # J/kg-K
    "density": 1200,              # kg/m^3 (approx. liquid phase)
    "latent_heat": 200000,        # J/kg
}
CU_PROPERTIES = {
    "thermal_conductivity": 401,  # W/m-K
}

# System Parameters
mass_flow_rate = 0.02  # kg/s (example)
refrigerant_inlet_temp = 40  # °C (for condenser)
refrigerant_exit_temp = 10   # °C (for evaporator)
ambient_temp = 25  # °C (for condenser surroundings)

# Heat Transfer Coefficient Calculation
def heat_transfer_coefficient(k, d):
    """Calculates the heat transfer coefficient."""
    return k / d

# Function to Calculate Required Pipe Length for Condenser
def calculate_condenser_length(diameter, heat_transfer_rate):
    """Calculates pipe length for the condenser."""
    h = heat_transfer_coefficient(R134A_PROPERTIES["thermal_conductivity"], diameter)
    delta_t = refrigerant_inlet_temp - ambient_temp
    length = heat_transfer_rate / (math.pi * diameter * h * delta_t)
    return length

# Function to Calculate Required Pipe Length for Evaporator
def calculate_evaporator_length(diameter, heat_transfer_rate):
    """Calculates pipe length for the evaporator."""
    h = heat_transfer_coefficient(R134A_PROPERTIES["thermal_conductivity"], diameter)
    delta_t = refrigerant_exit_temp - ambient_temp
    length = heat_transfer_rate / (math.pi * diameter * h * delta_t)
    return length
```

```
# Iterative Optimization
def optimize_pipe_dimensions(heat_transfer_rate, component):
    """Iteratively finds the optimal pipe dimensions."""
    diameter = 0.005  # Initial guess for diameter (m)
    for _ in range(100):  # Maximum 100 iterations
        if component == "condenser":
            length = calculate_condenser_length(diameter, heat_transfer_rate)
        elif component == "evaporator":
            length = calculate_evaporator_length(diameter, heat_transfer_rate)
        else:
            raise ValueError("Invalid component type")

        if length <= 2:  # Target length (example: 2 meters max)
            break
        diameter += 0.001  # Increment diameter (1 mm)

    return diameter, length

# Heat Transfer Rates (Assuming simplified values)
heat_transfer_rate_condenser = mass_flow_rate * R134A_PROPERTIES["latent_heat"]  # W
heat_transfer_rate_evaporator = mass_flow_rate * R134A_PROPERTIES["latent_heat"]  #
W

# Optimize Dimensions
condenser_diameter, condenser_length =
optimize_pipe_dimensions(heat_transfer_rate_condenser, "condenser")
evaporator_diameter, evaporator_length =
optimize_pipe_dimensions(heat_transfer_rate_evaporator, "evaporator")

# Results
print("Condenser Dimensions:")
print(f"  Diameter: {condenser_diameter:.4f} m")
print(f"  Length: {condenser_length:.2f} m")

print("\nEvaporator Dimensions:")
print(f"  Diameter: {evaporator_diameter:.4f} m")
print(f"  Length: {evaporator_length:.2f} m")
```

5.2 Mathematical Modelling

In this project, the evaporator and condenser tube consist of cylindrical thin tubes in which the refrigerant fluid passes that is mechanically supported by a mounting frame.

To mathematically model such a complex device, we assume that the flow rates of the fluids are constant. We also accept that the inlet temperatures of the fluid are unchanging. Moreover, it is taken for granted that the geometric structure and property of both condenser and evaporator tube material are homogenous.

However, to calculate this term, we use empirical correlations as follows:

For the tube side calculations. We use the Dittus-Boelter Equation

$$Nu = 0.023\, Re^{0.8} Pr^{n}$$

n=0.4 for heating and 0.3 for cooling.

Convection heat transfer rate for the tube side is calculated using the following formula

$$h = \frac{NuD}{k}$$

$$Pr = \frac{C_p \mu}{k}$$

$$Re = \frac{\rho V D}{\mu}$$

To calculate the equivalent heat transfer coefficient of the heat exchanger. As the tube is very thin, we assumed the inner and outer side area of the heat exchanger tubes are the same.

$$\frac{1}{h_{eqv.}} = \frac{1}{h_f} + \frac{t}{k} + \frac{1}{h_\infty}$$

Heat transfer and mass flow rate are calculated using the following formula

$$Q = hA\Delta T$$

$$m = \rho A V$$

Coefficient of Performance of the device are calculated using the following equation

$$CoP = \frac{Q_{evp.}}{W_{cond.}}$$

5.3 Calculations and Results

5.3.1 Calculation of Evaporator Section

$Q = 150\,W$

$T_{\text{evaporator}} = 5°\text{C}$

$T_{\infty} = 50°\text{C}$

$\triangle \text{T} = 45°\text{C}$

$\text{t} = 1\ \text{mm}$

$\text{K} = 385\ W/\text{m}K$

$h_{\infty} = 40\ W/\text{m}^2K$

$h_{\text{fluid}} = 25\ W/\text{m}^2K$

$$\frac{1}{h_{\text{eq}}} = \frac{1}{h_{\text{fluid}}} + \frac{t}{k} + \frac{1}{h_{\infty}}$$

$$\therefore \frac{1}{h_{\text{eq}}} = \frac{1}{25} + \frac{10^{-3}}{385} + \frac{1}{40}$$

$\therefore h_{\text{eq}} = 15\ W/\text{m}^2K$

$Q = h\,A \triangle \text{T}$

Here

$A = \pi DL$

$D = 9.5\ \text{mm}$

Solving this equation, we get, $L = 5.56\ \text{m}$

5.3.2 Calculation for Condenser Section

COP =3
$\dot{Q}_{cond} = 200$ W
Temperature of air, $T = 30°C$

$$h = 25 w/m^2 K$$

We use Refrigerant R- 134A by

Normally condenser length is half of the evaporator length.

So, $L = 2.78$ m
$D = 9.5$ mm

[we take the same diameter of pipe for evaporator tube and Condenser tube]
$T_{inner} = 50° C$

$T_{air} = 30° C$

$\Delta T = 20° C$

By using formula

$$\dot{Q} = hA\Delta T$$
$$200 = h \times (\pi \times 9.5 \times 10^{-3} \times 2.78) \times 20$$
$$h = 120.52 \text{ W/m}^2 \text{ K}$$

$$Nu = \frac{hD}{k} = \frac{120.52 \times 9.5 \times 10^{-3}}{0.08} = 14.31$$
$$P_r = \frac{C_p \mu}{k} = \frac{1.43 \times 10^3 \times 208 \times 10^{-6}}{0.08} = 3.718$$

We use This Correlation to Figure out Re numbers:

$$Nu = 0.023 Re^{0.8} Pr^{0.3}$$
$$14.31 = 0.023 \times Re^{0.8} \times (3.718)^{0.3}$$
$$Re = 1898.99$$
$$\frac{\rho VD}{\mu} = 1898.99$$
$$V = 0.0349 \text{ ms}^{-1}$$

Mass flow rate:

$$\dot{m} = \rho VA$$
$$= 1187 \times 0.0349 \times \frac{\pi}{4} \times (9.5 \times 10^{-3})^2$$
$$= 2.93 \times 10^{-3} \text{ kg/s}$$
$$= 2.93 \text{ g/s}$$

5.3.3 Calculation for Expansion Valve

Thermostatic Expansion Valve
Flow Capacity: $10 - 12$ kg/h
orifice size: $0.5 - 1$ mm
Refrigerant: $R - 134A$

5.4 Rating of the fabricated model

Heat Duty	150W
COP	3
Length per pass of the evaporator	9 inch

Chapter 6: Fabrication Method

6.1 Process of Fabrication

1. Preparation of the Materials and the Components
The fabrication started with choosing R-134A refrigerant, as far as its thermal characteristics and influence on the environment. A 50W compressor was purchased so as to provide the adequate cooling power for the laboratory. To provide a well regulated flow of refrigerant; a pre-calibrated expansion valve was bought. Condenser and evaporator required copper tubes for the fabrication process which was made according to the above design data. Heat transfer mechanism involving the condenser and the evaporator was isolated with the assistance of insulation material while an outer shell was arranged to cover all the elements snugly.

2. Fabrication of the Condenser
The copper tubing was coiled in a compact fashion so as to create something that is compact and yet occupying as much space as to expose a lot of surface area on the coils so as to be able to cool the fluid that flows through them. The tubing was thus fitted by joining it to the refrigerant inlet and outlet access points. All the joints were welded to prevent leakage where connections were made and to give the structure strength. Last, the condenser was tested for pressure to confirm its performance against operating pressure.

3. Fabrication of the Evaporator
The evaporator was built from the copper tubing shaped in a way which creates direct contact with the laptop surface necessary for cooling. Tubing was arranged to the refrigerant flow system, and all joining areas were properly welded in order to avoid leakage. To validate the successful distribution of heat at the evaporator and its physical strength the thermal test was conducted by heating the metal plate up to certain intervals.

4. Integration of Components
The expansion valve was then fitted in-between the condenser and evaporator to ensure it is well fixed and in the right position. The 50W compressor was installed inside the outer shell leaving adequate room for airflow and wiring harnesses. This cooling system included condenser, evaporator, compressor, and the expansion valve functioning a closed refrigerant circuit made with a copper tube. All joints were made by welding so that no leakages were possible between one section and the next.

5. Insulation and Thermal Separation
Economically, insulation and thermal separation represent a key element as a barrier between two different temperature areas. High-quality insulation material was used between a

condenser and an evaporator to avoid energy loss due to heat transfer between hot and cold surfaces. This tubing was then exposed in order to be insulated so as to reduce loss of energy and enhance efficient use of the tubing.

6. Placement into Outer Shell
All were enclosed in a strong outer casing so nothing could be damaged or come loose during operation. Each condenser and evaporator was arranged to have the right position and air flow direction. Its core components which include the clipper and the tubing were secured in the shell by clamps as well as brackets to minimize on movement during functioning. They made some openings in the shell to ensure that there is a free flow of air around the condenser.

7. Refrigerant Charging
It was then serviced through a service port charged with R-134A refrigerant. Flow and pressure were regulated in the process to achieve adequate refrigerant pressure.

8. Testing and Quality Assurance
The unit's leak testing was done by infusion of a vacuum pump and pressure gauges to confirm no refrigerant leaks. The cooling pad was used under different loads to confirm its ability in lowering laptop temperatures. The temperatures at condenser and evaporator was taken with the aim of verifying the effectiveness of the refrigeration cycle.

9. Final Assembly and Finishing
Body sections and refrigerant circuits were pre-marketered with a protective outer casing. For stability extra non-slip pads were placed underneath the base. A final operational check was conducted to check whether the cooling pad itself is up to the design requirement and whether it gives a good cooling effect.

This fabrication process assures safe and successful assembling of the laptop cooling pad as well as efficiency of the product. Should you require any finer alterations or more specific information do not hesitate to contact me.

6.2 Costing

No.	Name Sector	Unit Cost (BDT)	Total Cost(BDT)
1	Evaporator	30(per ft)	6480
2	Condenser Tube	24(per ft)	2600
3	Expansion Valve		650
4	Box		400
5	Fittings		320
6	Labor		1500
7	Transports		400
8	Others		150
Total			12,500

6.3: Flow Circuit Testing

After making the prototype, it was taken to BUET Fluid Lab to Flow circuit test. The project successfully passed the flow circuit test.

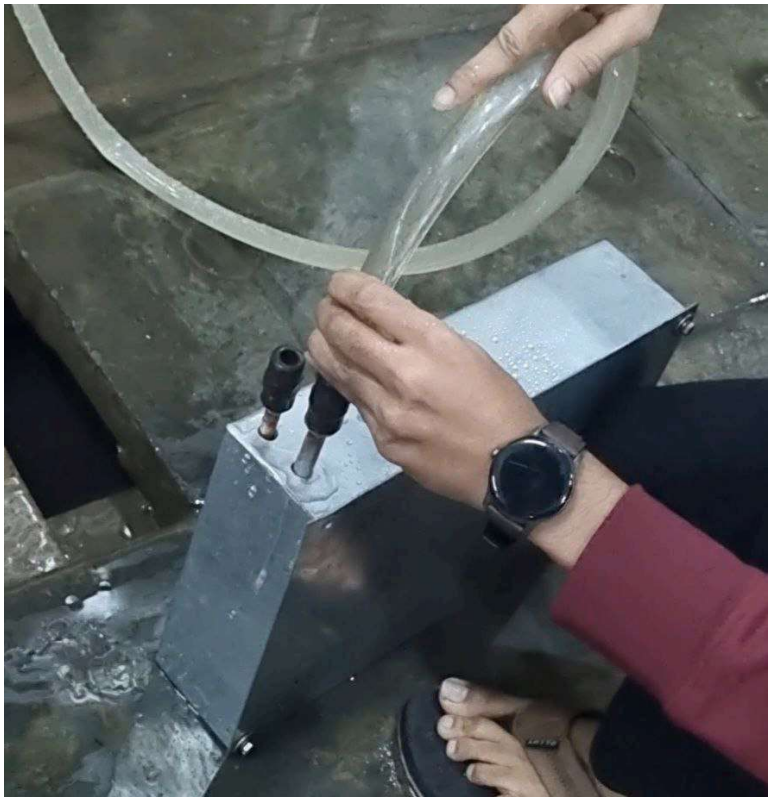


Fig: Setup for Flow Circuit Test

Bangladesh University of Engineering & Technology

Course no: ME-310

Course Title: Thermo-Fluid System Design

Project Title: **Portable Cooling Pad: A refrigeration Cycle for Laptop**

Submitted by-

Group 02, Section: A1

2010004 – Juthi Hira

2010011 – Amet Hasan Emon

2010015 – Fayez Uddin Ahmed Farhan

2010017 – Avijit Mondal Shantonu

2010029 – Kishar Das Gupta

This is to certify that '**Portable Cooling Pad: A refrigeration Cycle for Laptop**' has successfully passed the Flow Test. No leakage was observed.

Received by Fluid Lab.

Date: 13.12.24

Time: 3:30 PM Sign.....

Signature

Scanned with CamScanner

6.4: Pressure Testing

After making the prototype, it was taken to BUET Hydraulics Lab for a pressure test. To test the pressure, the outlet was closed by a death cap after the tubes were filled with water and then in the inlet water pressure was being increased using a hand pump. And the amount of pressure given was seen from the pressure gauge attached with the hand pump. And the lab officials confirmed before the test, that the pressure gauge was calibrated properly. And the pressure given is slightly greater than 200 psi . So, it's confirmed that our prototype can withstand at least a pressure of 200 psi. With more safety measurements, and other equipment may be checked the maximum pressure the prototype can withstand before failing. From the pressure testing, we can come to the conclusion that the selection of materials and joining process was good enough.



Fig: More than 200 psi Pressure can be sustained

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This is to certify that **'Portable Cooling Pad: A refrigeration Cycle for Laptop'** has successfully passed the Hydraulic Test. No leakage was observed.

(200 Psi sustained)

তিথি

তিথি 17/12/24

Lab Instructor, ME, Buet

Signature

CS Scanned with CamScanner

Conclusion

The creation of a refrigeration cycle laptop cooler is a prime example of progress in thermal control of portable electronic appliances. Through the technology of Cu tubes and R-134A refrigerant, the project effectively caters to the problems existing with fan-based or passive cooling systems. The cycles of design enabled the correct dimensions of the condenser and the evaporator to be determined so that the heat transfer system could be made to be compact and functional at the same time.

There is always a maximum requirement of using environmentally sound refrigerant which R-134A fulfils while the use of copper tubes increases conductivity and product life. Now, owing to this novel procedure, not only does Wraith Prism enhance cooling for high-end laptops, but also their longevity, as constant overheating hastens laptop deterioration, especially during heavy operations, like gaming, video designing or simulation engineering. This project demonstrates that students beginning with relatively basic concepts in thermodynamics can apply them to solve a devise practical problem. Possible avenues for future research include the further miniaturization of the system, investigation into more environment friendly refrigerants with a global warming potential even lesser than the one used in the system, and energy efficiency improvement. With these advancements the concept of cooling based refrigeration cycle will be a pathway towards discovering more efficient and environment friendly thermal management solutions in portable electronics.

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